CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 18: Statistics

Topic-wise short notes with solved examples

Collecting and organising data: variables, frequency distributions, class intervals and their limits, cumulative frequency, and the graphs used to show grouped data (histogram and frequency polygon).

1 Data

- **Statistics** deals with collecting, organising, presenting and interpreting numerical facts.
- **Data** are the facts collected. **Primary data** are collected by the investigator; **secondary data** are taken from someone else's records.
- **Raw data** are data as first collected. Arranged in ascending or descending order, they form an **array**.
- A **variable** (variate) is a quantity that is measured. A **discrete** variable takes separate values (number of children); a **continuous** variable can take any value in a range (height, mass).
- **Range** = greatest value − least value.
- **Frequency** of a value: the number of times it occurs.

2 Frequency distribution

- A table of values with their frequencies is a **frequency distribution**. Use **tally marks** to count: four strokes and a fifth across them.
- **Ungrouped**: each value listed separately. **Grouped**: values put into **class intervals**.

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	3	7	12	6	2

- The total of the frequencies equals the number of observations (here 30).

3 Class intervals

- **Class limits**: the two end values. In 10–20, the **lower limit** is 10 and the **upper limit** is 20.
- **Exclusive (continuous) form**: 0–10, 10–20, 20–30. The upper limit of a class is excluded from it: 10 goes into 10–20.
- **Inclusive (discontinuous) form**: 1–10, 11–20, 21–30. Both limits belong to the class.
- **Class size (width)** = upper boundary − lower boundary.
- **Class mark** (mid-value) $= \dfrac{\text{lower limit} + \text{upper limit}}{2}$.
- **Converting inclusive to exclusive**: find the gap between the upper limit of one class and the lower limit of the next. The **adjustment factor** is half the gap. Subtract it from every lower limit and add it to every upper limit. The results are the **true class limits** (class boundaries).

Example 3.1 Convert the classes 1–10, 11–20, 21–30 to exclusive form. Find the class size and the class marks.

SOLUTION

Gap $= 11 - 10 = 1$, so the adjustment factor is 0.5.

0.5–10.5, 10.5–20.5, 20.5–30.5; class size 10; class marks 5.5, 15.5, 25.5

Example 3.2 The class marks of a distribution are 12, 18, 24, 30. Find the class size and the class intervals.

SOLUTION

Class size = difference of consecutive class marks $= 6$. Each class runs 3 below and 3 above its mark.

9–15, 15–21, 21–27, 27–33

4 Making a grouped table

- Find the range. Choose a class size so that there are about 5 to 10 classes.
- Go through the data once, putting a tally mark against the class of each value.
- Count the tallies and check that the frequencies add up to the number of observations.

Example 4.1 Make a frequency table with classes 20–30, 30–40, 40–50, 50–60 for the marks: 23, 45, 38, 52, 41, 29, 34, 47, 55, 31, 26, 44, 39, 50, 36.

SOLUTION

20–30: 23, 29, 26. 30–40: 38, 34, 31, 39, 36. 40–50: 45, 41, 47, 44. 50–60: 52, 55, 50.

Frequencies 3, 5, 4, 3 (total 15)

5 Cumulative frequency

- The **cumulative frequency** of a class is the sum of the frequencies of that class and all the classes before it.
- The last cumulative frequency equals the total frequency.
- To get the frequencies back, subtract consecutive cumulative frequencies.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	3	7	12	6	2
Cumulative frequency	3	10	22	28	30

Example 5.1 From the table, how many students scored less than 30? How many scored 20 or more?

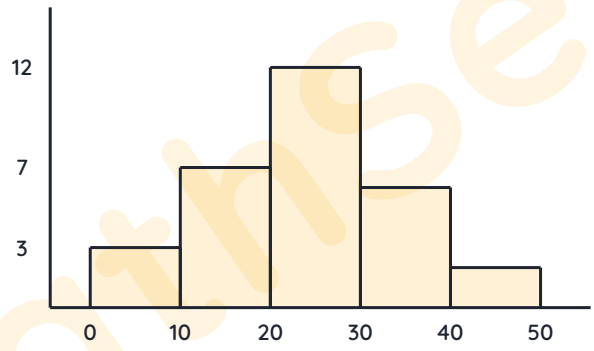
SOLUTION

Less than 30: cumulative frequency of 20–30 is 22. 20 or more: $30 - 10$.

22 and 20

6 Histogram

- A **histogram** shows a continuous grouped distribution by **adjacent rectangles**. Class intervals are on the horizontal axis and frequencies on the vertical axis.
- With equal class sizes, the heights are proportional to the frequencies. There are **no gaps** between the rectangles.
- If the classes are inclusive, change them to exclusive form first.
- If the scale does not start at zero, show a **kink** (zig-zag) on the axis.



Histogram of the marks table

7 Frequency polygon

- A **frequency polygon** is obtained by joining, with straight lines, the points plotted at (class mark, frequency).
- Close it by adding one class of zero frequency before the first class and one after the last.
- **From a histogram**: mark the mid-points of the tops of the rectangles and join them in order.
- Frequency polygons are useful for comparing two distributions on the same axes.

Example 7.1 For the marks table above, write the points to be plotted for the frequency polygon.

SOLUTION

Class marks 5, 15, 25, 35, 45 with frequencies 3, 7, 12, 6, 2. Imagined classes -10–0 and 50–60 have class marks -5 and 55.

$(-5, 0), (5, 3), (15, 7), (25, 12), (35, 6), (45, 2), (55, 0)$

8 Common mistakes

- Putting a value equal to an upper limit (such as 30 in 20–30) in the lower class in the exclusive form. It belongs to 30–40.
- Drawing a histogram for inclusive classes without adjusting the limits, which leaves gaps.
- Confusing a histogram with a bar graph. A bar graph has gaps and equal-width bars for separate items.
- Plotting a frequency polygon at class limits instead of class marks.
- Not closing the frequency polygon on the horizontal axis.
- Forgetting to check that the frequencies add up to the total.

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